Jinwon Lee

Platform & Site Reliability Engineer

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PROFILE

Platform and site reliability engineer with about eight years of experience building and operating cloud-native application platforms across AWS EKS, Azure, Kubernetes, and OpenShift.

Combines platform software, release automation, observability, access control, and incident response to move services into reliable production operation.

SELECTED EVIDENCE

AWS EKS agent platform with automated domains and authorized-only access	Production delivery and reliability for a 30,000-employee AI service	Kubernetes delivery, observability, and incident response across cloud environments

CORE SKILLS

Platform	AWS EKS, Kubernetes, OpenShift, Azure, Docker, Helm/Helmfile, Kubernetes Operators
Delivery Automation	ArgoCD, GitOps, GitHub Actions, Jenkins, CI/CD, Bash-based cluster provisioning
Reliability	Grafana, Prometheus, OpenTelemetry, high availability, auto-scaling, incident response
Software & Security	Python, Bash, Linux, API integration, authentication, authorization, security governance

EXPERIENCE

Samsung Electronics, AX Development Group
Seoul, South Korea
SRE / AI Platform Engineer — Jan. 2026 - Present

- Design and build an internal AX Agent Platform on AWS EKS, adapting the agent runtime and deployment path to enterprise platform standards.
- Implemented self-service AX application delivery with automatic domain issuance, authorized-only access, and platform controls for authentication, authorization, security, and governance.
- Enabled AX applications to call the existing enterprise LLM Gateway API through platform-side client compatibility and runtime integration.
- Supported a Microsoft-collaborative B2C generative AI service through UK rollout readiness, initial production operations, incident response, and stabilization from Jan. to Jun. 2026.

Samsung Research
Seoul, South Korea
SRE / AI Platform Engineer — Jul. 2024 - Dec. 2025

- Led production delivery and application reliability for a company-wide AI chatbot serving 30,000 employees on internal Kubernetes clusters.
- Owned releases across development, staging, and production while integrating application delivery with security and operational requirements.
- Built container delivery pipelines with Docker, Kubernetes, GitHub Actions, ArgoCD, Helm, and Helmfile.
- Deployed Grafana, Prometheus, and OpenTelemetry observability and built a Python/MLflow evaluation workflow for LLM-based machine translation.

Samsung Electronics, Network Division
Seoul, South Korea
SRE & Software Engineer — Jan. 2021 - Jun. 2024

- Designed and delivered containerized applications with Docker, Helm, Kubernetes Operators, and AWS services including EKS, ECS, Route 53, RDS, and IAM.
- Built and managed OpenShift, Kubernetes, and EKS environments for reliability, high availability, performance, and auto-scaling.
- Designed and developed a Bash-based one-click Kubernetes cluster provisioning system for cloud developers.
- Developed a Linux application that mirrors and forwards up to 130,000 packets per second under constrained system resources.

EXPERIENCE (CONTINUED)

TmaxOS & TmaxCloud
Gyeonggi, South Korea

Software & Platform Engineer
Jan. 2019 - Dec. 2020

- Built a Siri-like AI assistant on TmaxOS, covering model inference, data preprocessing, system integration, and asynchronous application flows.
- Designed and developed Kubernetes CI/CD and registry operators and deployed a private container image registry.

SELECTED ENGINEERING WORK

Kubernetes Cluster Automation
Designed and developed a Bash-based one-click provisioning system for repeatable developer Kubernetes environments.

Centralized Observability Stack
Deployed Grafana, Prometheus, and OpenTelemetry for shared service visibility, dashboards, and operational diagnosis.

High-throughput Packet Mirroring
Developed a resource-aware Linux application measured at up to 130,000 packets per second.

Internal AX Agent Platform
Built an AWS EKS application platform with automatic domains, authorized-only access, enterprise security controls, and compatibility with an existing LLM Gateway API.

EDUCATION & RESEARCH

UNIST | M.S. in Computer Science and Engineering
Mar. 2017 - Feb. 2019

Master's thesis: Convergence Aware CNN Training | Advisors: Jiwon Seo and Sam H. Noh

UNIST | B.S. in Computer Science and Engineering, Minor in Mathematics
Mar. 2012 - Feb. 2017

Undergraduate research: distributed systems for AI model training with TensorFlow

PUBLICATIONS & LANGUAGE

- USENIX NSDI 2019 poster: Alleviating the Network Bottleneck for CNN Distributed Training through Automatic Resource-Aware Layer Placement.
- arXiv 2019: Accelerated Training for CNN Distributed Deep Learning through Automatic Resource-Aware Layer Placement.
- ACM EuroSys 2018 poster: Improving Performance of Distributed TensorFlow using CNN Characteristics Exploiting Model Parallelism.
- OPIc IH (Intermediate High), Dec. 2025.